UEI613: BIOMETRICS

L	T	P	Cr
2	0	2	3.0

Course Objectives: To understand the concepts of Biometrics and to design biometric system

Introduction: Overview of Biometrics, Biometric Identification, Biometric Verification, Biometric Enrollment, Biometric, System Security, Introduction of biometric traits and its aim, image processing basics, basic image operations, filtering, enhancement, sharpening, edge detection, smoothening, enhancement, thresholding, localization. Fourier Series, DFT, inverse of DFT

Authentication and Biometrics: Secure Authentication Protocols, Authentication Protocols, Biometric system, identification and verification. FAR/FRR, system design issues. Positive/negative identification. Biometric system security, authentication protocols, matching score distribution, ROC curve, DET curve, FAR/FRR curve. Expected overall error, EER, biometric myths and misrepresentations.

Common biometrics: Finger Print Recognition, Face Recognition, Speaker Recognition, Iris Recognition, Hand Geometry, Signature Verification, Positive and Negative of Biometrics.

Selection of suitable biometric: Biometric attributes, Zephyr charts, types of multi biometrics. Verification on multimodel system, normalization strategy, Fusion methods, Multimodel identification.

Course Learning Outcomes (CLO):

Student will be able to:

1. Elucidate the basics of Biometric Identification
2. Analyze different error measures used in biometric identification and verification
3. Apply various biometrics for identification
4. Exhibit the knowledge of multi-model biometric identification system.

Text Books:

1. *Digital Image Processing using MATLAB, By: Rafael C. Gonzalez, Richard Eugene Woods, 2nd Edition, Tata McGraw-Hill Education 2010*
2. *Guide to Biometrics, By: Ruud M. Bolle, Sharath Pankanti, Nalini K. Ratha, Andrew W. Senior, Jonathan H. Connell, Springer 2009*
3. *Pattern Classification, By: Richard O. Duda, David G.Stork, Peter E. Hart, Wiley 2007*

Reference Books:

1. *Bolle, Connell et. al., "Guide to Biometrics", Springer.*

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1	MST+EST	70
2	Sessional (May include Assignments//Quizzes/Lab Evaluations)	30

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 114th meeting of the Senate held on March 11, 2024 and March 7, 2025 respectively.